Sadashiba Sarangi

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Summary

Cybersecurity-focused Software Developer with expertise in Application Security (AppSec), Vulnerability Assessment, Penetration Testing, and SSDLC. 2x Bronze Medalist at Odisha Skills (Cyber Security). Experienced in LLM-powered security automation, mobile penetration testing, RAG systems, and full-stack web development. Proficient in Python, TypeScript, and JavaScript with hands-on Linux administration, network security, and OWASP compliance.

Education

C.V. Raman Global University — Bhubaneswar, Odisha
Bachelor of Technology (B.Tech) – Computer Science and Information Technology — *2023 – Present*

C.V. Raman Polytechnic — Bhubaneswar, Odisha
Diploma – Computer Science Engineering — *August 2019 – April 2023*

Technical Skills

Programming Languages: Python, TypeScript, JavaScript, Java, Bash, SQL
Security Tools: Wireshark, Metasploit, OWASP ZAP, Nmap, OpenVAS, Burp Suite, Frida, ADB
AI and Machine Learning: OpenAI API, Google Gemini API, TensorFlow, OpenAI Whisper, Ollama, LangChain, RAG
Web Development: Next.js 14, React 18, Node.js, Express.js, FastAPI, Tailwind CSS, HTML, CSS, REST APIs
Databases and DevOps: PostgreSQL, MongoDB, Docker, GitHub Actions, Git, CI/CD
Operating Systems: Kali Linux, Ubuntu, Windows

Projects

LLM-Based Autonomous Mobile App Security Penetration Testing Agent 2025 – 2026

Personal Project | Python, FastAPI, Frida, Ollama, RAG | Source Code

- Architected a 10-phase autonomous security scanning pipeline for Android APK vulnerability detection using Python, FastAPI, and Frida, achieving a **94.2% detection rate** with only **7.8% false positives**.
- Engineered a Retrieval-Augmented Generation (RAG) engine indexing **2,075 security documents** (OWASP, CWE, CAPEC, NVD) with BM25 and TF-IDF hybrid retrieval and Reciprocal Rank Fusion for context-grounded LLM reasoning.
- Integrated Ollama-hosted Qwen 2.5-Coder LLM (1.5B parameters) for vulnerability classification, false-positive filtering, and automated remediation code patch generation.
- Implemented dynamic runtime analysis using Frida instrumentation and ADB for hooking Android API calls; achieved full **OWASP Mobile Top 10 (2024)** and **OWASP API Top 10 (2023)** coverage across **104 real-world applications**.
- Designed a 100% privacy-preserving architecture with zero cloud dependency – all LLM inference and data processing executed locally via Ollama.

KSecurity – AI-Powered Community Threat Intelligence Platform 2025

Personal Project | TypeScript, Next.js 14, Node.js, OpenAI API, PostgreSQL, Docker | Source Code

- Built a real-time AI-powered threat detection platform protecting Discord communities from phishing and social engineering using three parallel analyzers (LLM classifier, URL phishing engine, behavioral tracker) with weighted composite risk scoring.
- Integrated OpenAI GPT-4o-mini for real-time message classification across 6 threat categories with a 3-second circuit breaker and automatic heuristic fallback for zero-downtime detection.
- Engineered a URL phishing analysis engine with typosquatting detection for 10+ brands and a behavioral anomaly tracker with Jaccard similarity for spam detection, burst monitoring, and account-age risk scoring.
- Built a Next.js 14 admin dashboard with JWT authentication, real-time threat analytics, and audit logs; architected a TypeScript monorepo with Docker Compose, Railway PaaS, and GitHub Actions CI/CD pipeline.

KrishiMitra – Multimodal AI-Powered Agricultural Assistant 2025

Personal Project | Python, FastAPI, Google Gemini, TensorFlow, Twilio, MongoDB | Source Code

- Led development of a multimodal AI chatbot deployed on WhatsApp (via Twilio) providing real-time agricultural advisory to Indian farmers, routing text, image, and audio to independent AI pipelines – Gemini for NLP, MobileNetV2 CNN (**94.7% accuracy** on 54,000+ images), and Whisper for speech-to-text.
- Engineered zero-shot prompt guardrails achieving **0% out-of-domain hallucination rate**; integrated weather APIs and a RAG-based knowledge retrieval engine for government scheme eligibility.
- Resolved production bottlenecks: mitigated Twilio 15-second webhook timeouts via async job queues, eliminated HTTP 429 rate-limit exhaustion with Redis caching layer, and tuned MongoDB connection pooling for high-concurrency throughput.

Certifications and Achievements

3rd Position (Bronze Medal) – Odisha Skills Competition 2025–26 — Cyber Security
3rd Position (Bronze Medal) – Odisha Skills Competition 2020–21 — Cyber Security

Additional Experience

- Web application exploitation: SQL Injection (SQLi), Cross-Site Scripting (XSS), Authentication Bypass in CTF competitions.
- Network forensics and PCAP analysis using Wireshark. Linux privilege escalation and system hardening labs.